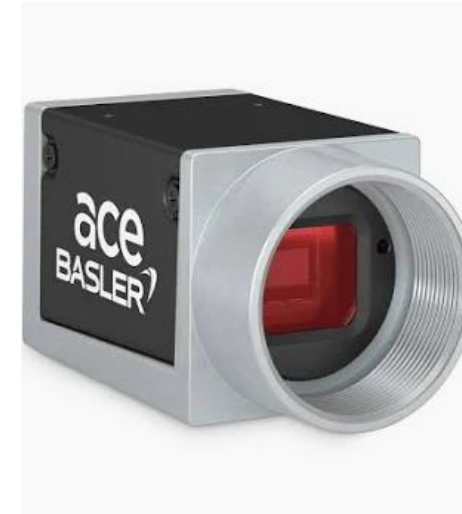


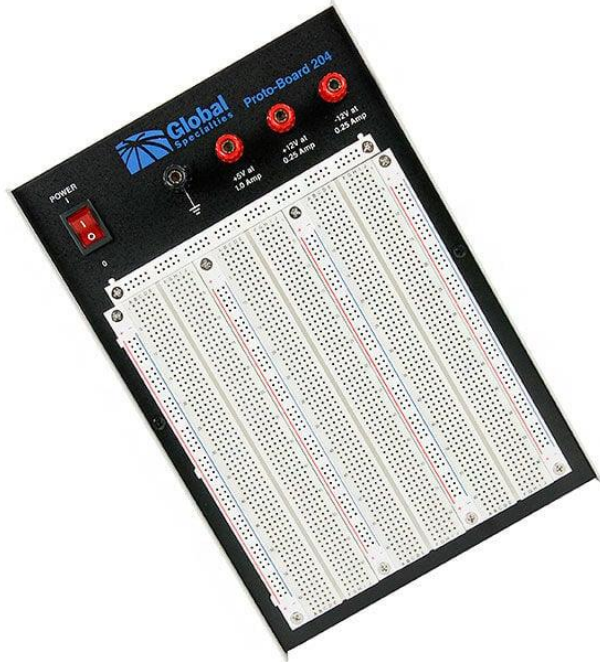
ENPH 334 Lab Overview

September 2026

Measurements mediated by electronics



Main goal of the labs: learn how to use the equipment



Breadboard: Build and test circuits with no soldering



Function generator: Create AC waveforms to send through your circuit

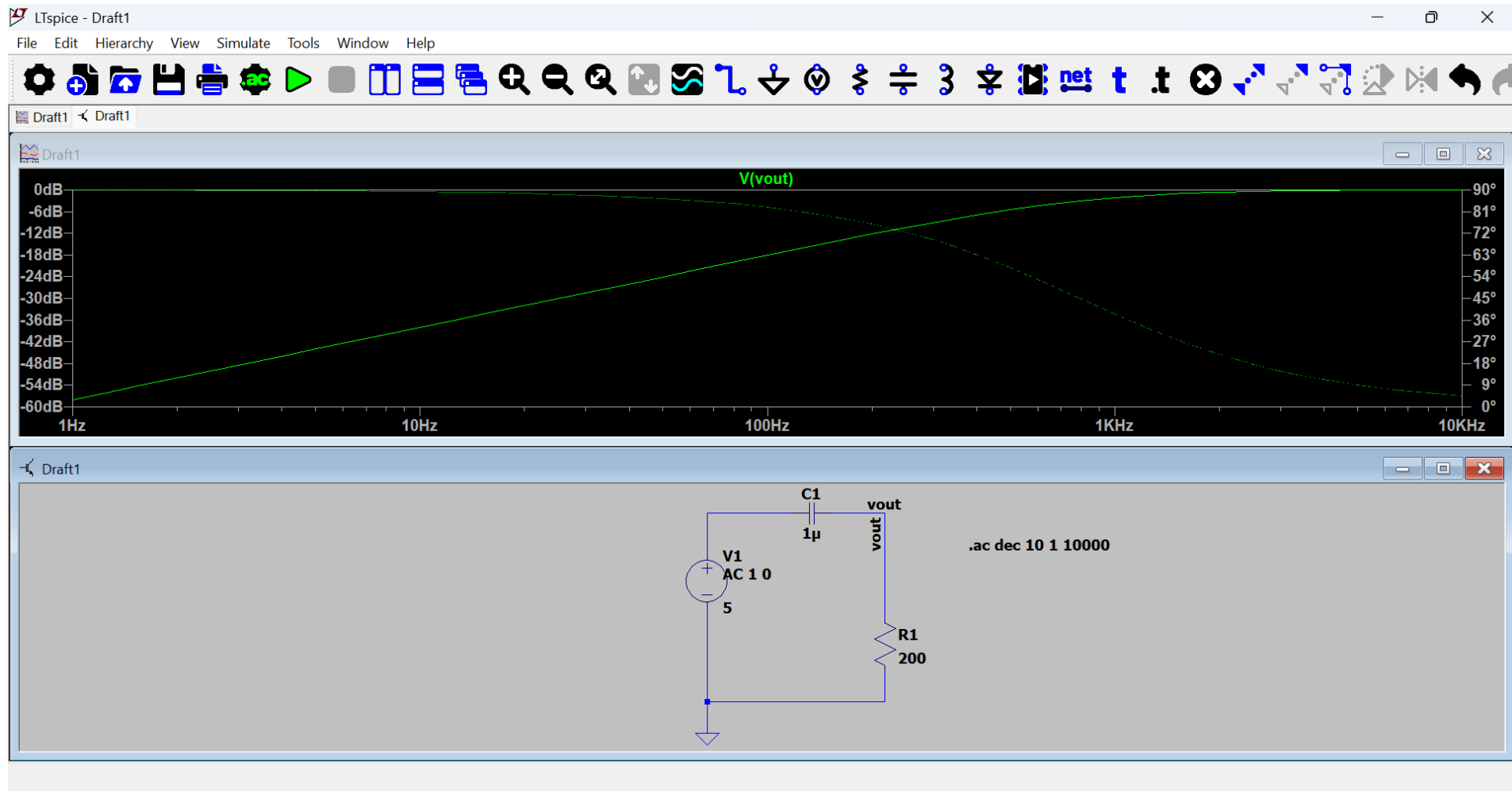


Multimeter: Measure voltage, current, resistance, and more



Oscilloscope: Measure voltage as a function of time!

Secondary goal: design and model electrical circuits



Outline

- Teaching team
- Lab structure
- Logistics
- How to succeed

Teaching team: lectures & “theory”

Instructor: Dr. Bhavin Shastri

- Lectures
- Course admin

TA: Arpan Sur

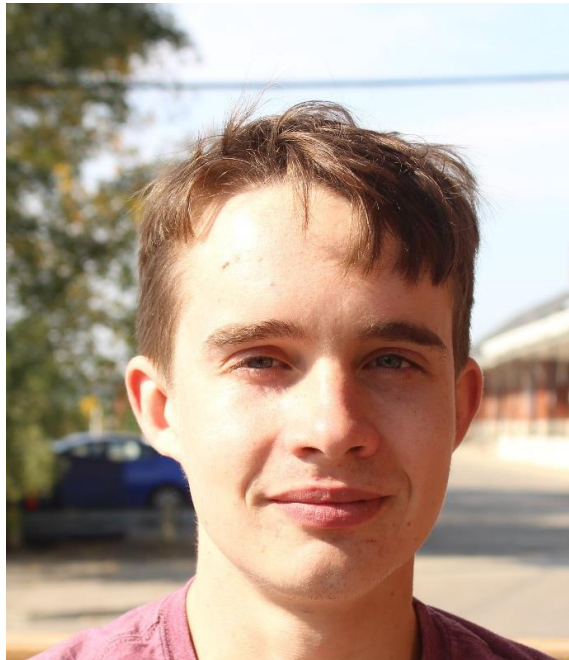
- Quizzes
- Exam



Teaching team: labs

Fraser McCauley

- Lab support
(Mon,
Wed morning)
- Lab admin



Sam Lamontagne

- Lab support
(Wed evening,
Thurs)



Ammar Ibrahim

- Lab support
(Mon,
Wed morning)
- Notebook grading



Cameron Ingo

- Lab support
(Wed evening,
Thurs)
- Notebook grading

Lab structure: Four components

Completion

Show up every week and complete the tasks in the lab manual.

Grading is binary (you must finish all lab tasks to get ANY credit on the lab portion of the course).

Lab notebooks

Document your work in the lab, *during your lab sessions*. These notebooks will be evaluated twice: once for feedback, once for grading.

Design projects

You will be asked to design, build and characterize two circuits: analog in Week 8, digital in Week 11. You will have to demonstrate functionality of your circuit to the TAs, answer questions about it, and submit a formal report.

Lab test

End of term. Individually evaluated on proper usage of the lab equipment. You will be asked to make simple measurements of circuits used in the labs.

Final Exam	40%
Quizzes (×4) best 3/4	21%
Design Projects (×2)	24%
Laboratory Test	8%
Laboratory Notebook	7%
Total	100%

Weekly lab completion

- Performed in groups of 2, formed at the first lab.
- Read the lab manual, perform calculations, etc. before attending the lab. Time during the lab is very limited.
- Labs are organized into **Tasks** which are the deliverables
 - Some experimental (ie. “measure the voltage”)
 - Some theoretical (ie. “calculate the capacitance required to...”)
 - Some simulated (ie. “use LTSpice to simulate the frequency response of...”)
- Simulation and theoretical tasks can be started/completed before the start of the lab, and checked off immediately

Lab notebooks

- Keep a continuous notebook documenting all the labs you complete
- Requirements in the “Course Information” document on OnQ
- Physical or digital notebooks are both fine, but must be uploaded in a legible PDF format.
- Submit the **first half** of your notebook at the end of Week 4.
 - **Not graded**, but TAs will give feedback to help with notetaking during later labs
- Submit **second half** of your notebook at the end of Week 11
 - **Graded**

Projects

- Two projects in the course: Analog and Digital
- These are design projects. You are expected to do research outside of what is presented in lecture notes.
 - Hint: check out the Resources slide
- Analog Project Presentation in Week 8
 - Week 7 has a work period dedicated to working on the project
- Digital Project Presentation in Week 11
 - There is no extra work period for this project.
- **Finish labs early so you can spend lab time on the projects!**

Logistics: lab sections

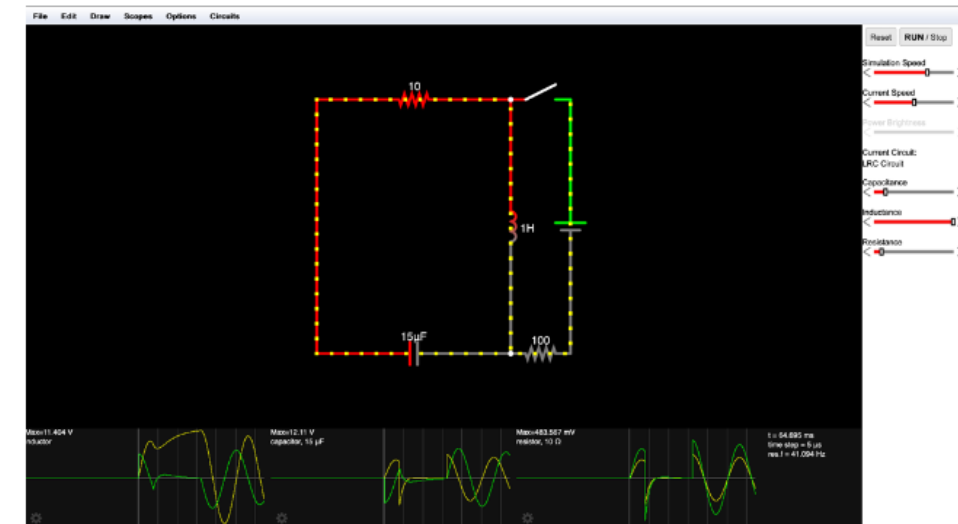
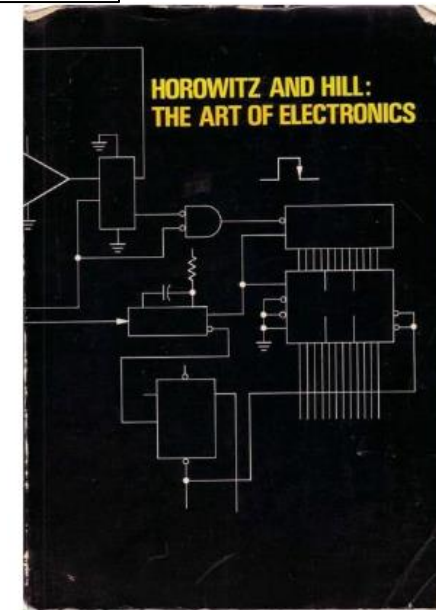
- All labs are in Stirling room 405
- **Attend the lab section you were assigned**
- Lab section swaps will **only** be permitted on a 1-to-1 basis
 - Find someone in your desired section who wants to switch into yours
- If you want to swap sections (and have identified someone to switch with), send me an email ASAP!
fraser.mccauley@queensu.ca

Logistics: completion tracking

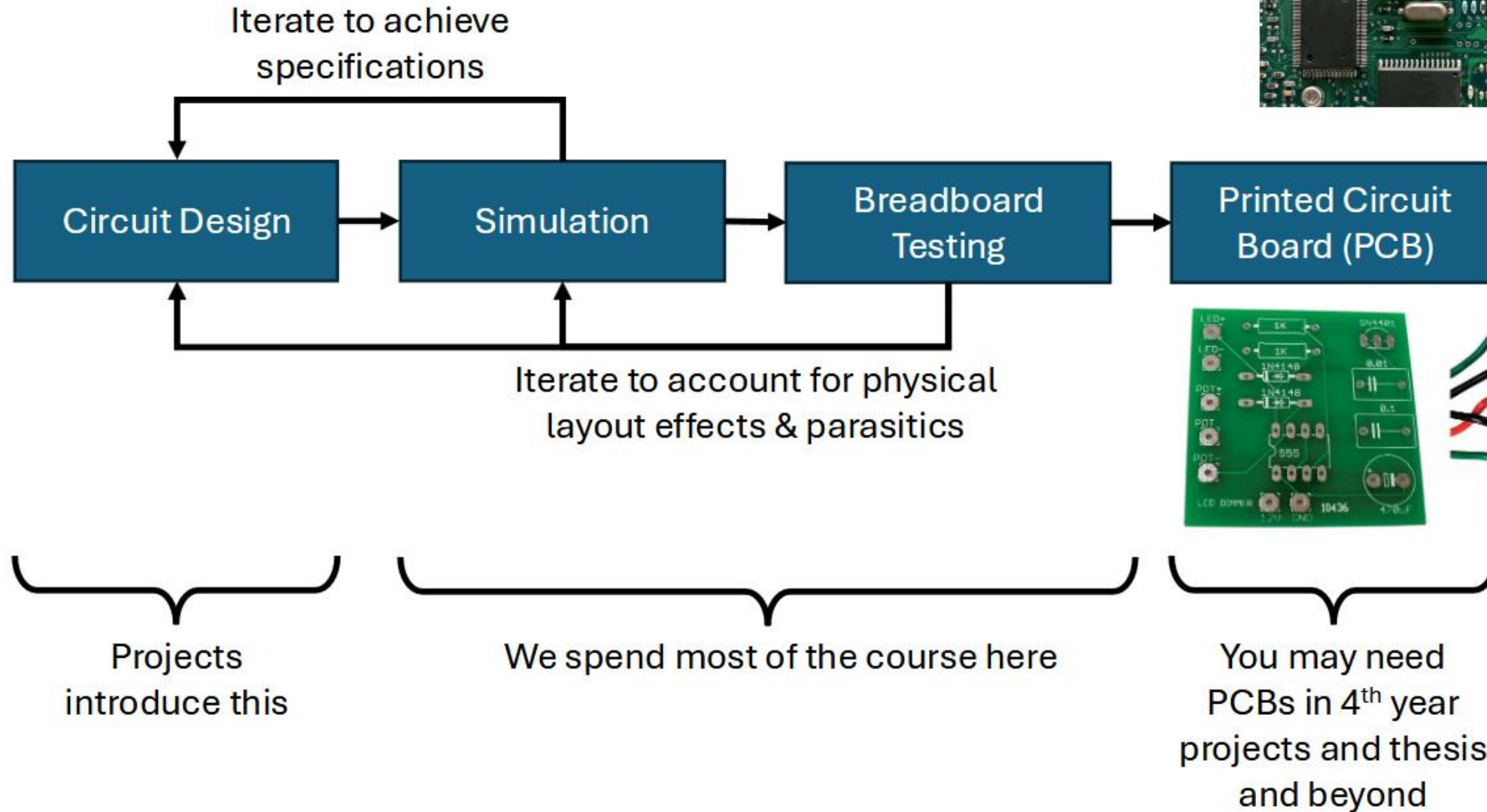
- When you complete a task in the lab manual, raise your hand and a TA will check that it is completed properly, and check you off for that task.
- You must complete **ALL** the labs to get credit for the “Lab Notebook” and “Design Project” components of the course (36%).
- If you miss a lab, you have a few options to make it up:
 - Finish at the start of your next session
 - Attend a different lab section to complete it (**email TAs before showing up**)
 - Attend outside of lab hours (talk to TA to be let in to the lab)
 - You need to coordinate around the TAs’ schedule
 - We won’t stick around for you to ask questions
 - Week 7 has an extra lab session to work on your analog project that you can use to make up a missed lab
- **It is your responsibility to make sure a TA has checked your completion before leaving the lab.**

Resources

- Alternative Textbook: *The Art of Electronics*, Horowitz & Hill
 - Not official for this course, but a great resource
 - One copy in the lab as a reference but it must stay in the lab
- Another alternative: *Principles and Applications of Electrical Engineering*, Rizzoni & Kearns
- Circuit Simulator:
<https://www.falstad.com/circuit/circuitjs.html>
 - Not as powerful as SPICE, but gives intuition for voltage and current flow
- Design Resources: Textbooks, application reports, technical notes are indispensable
 - Odds are that you aren't the first person to design a circuit!
 - Any PDF released by Texas Instruments is amazing
 - Online resources like "electronics-tutorials.ws" can be helpful, but they don't tell the complete story



Electronics Design Flow (simplified)

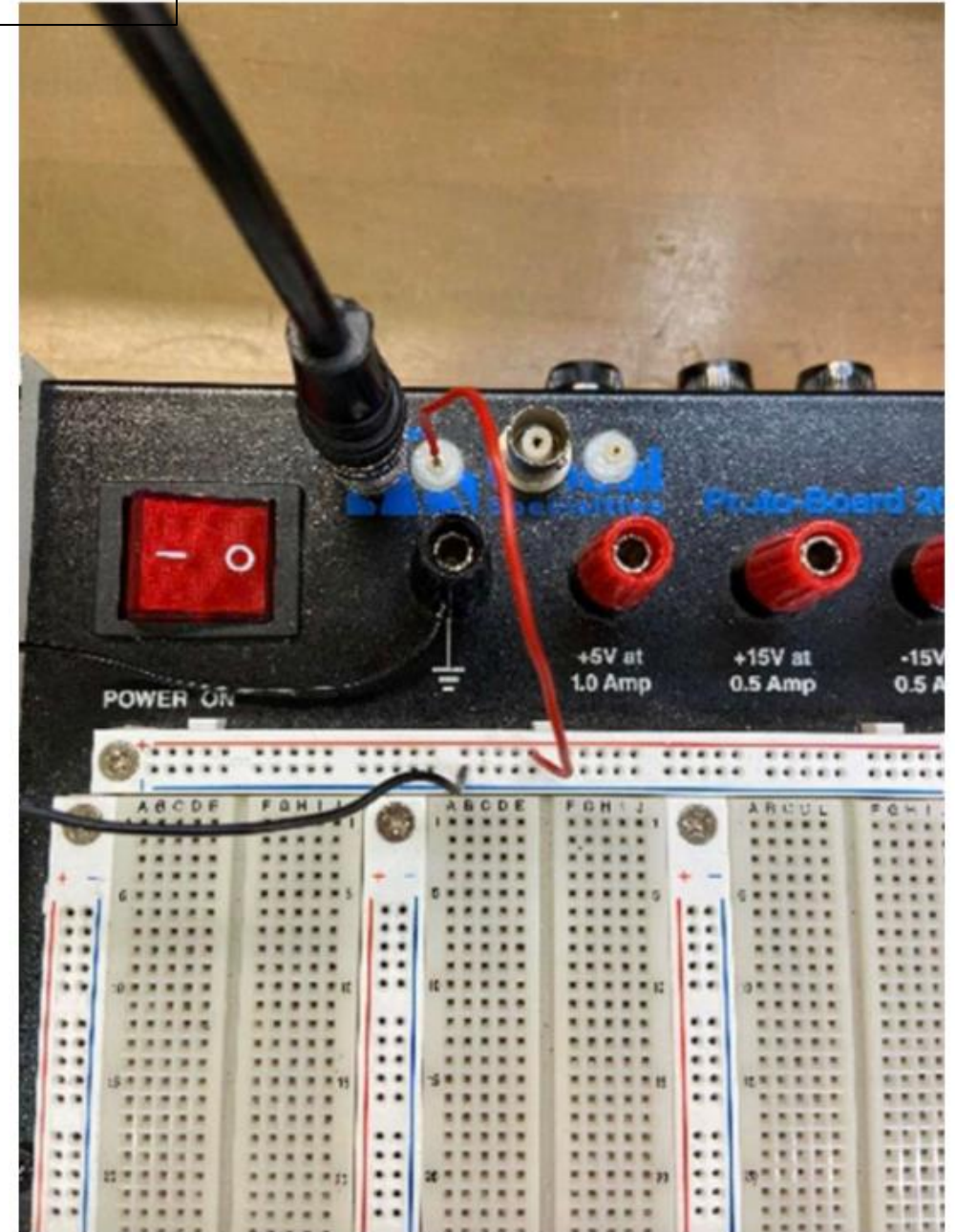


Practical Electronics

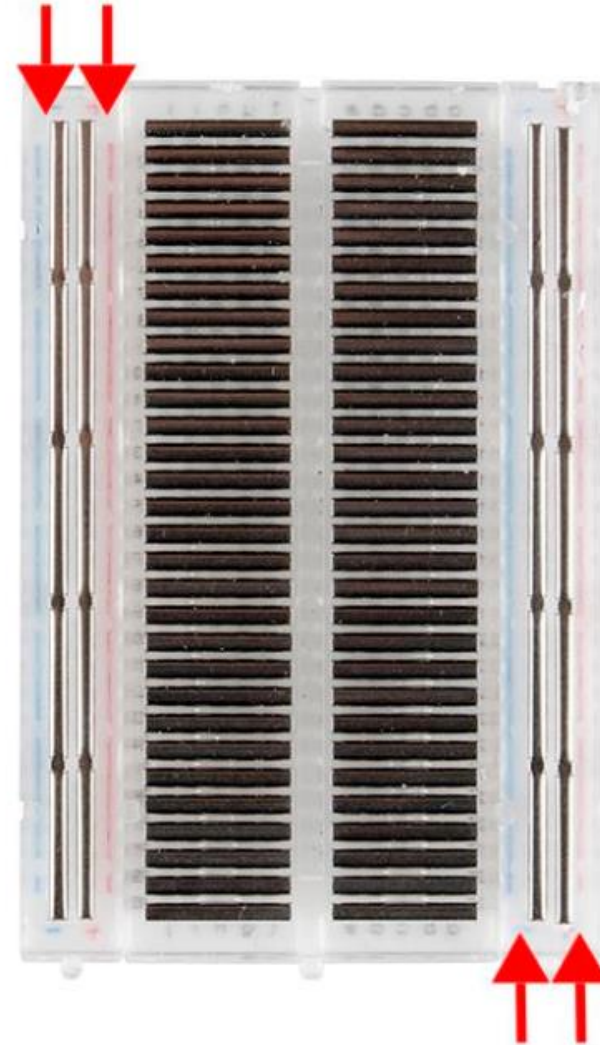
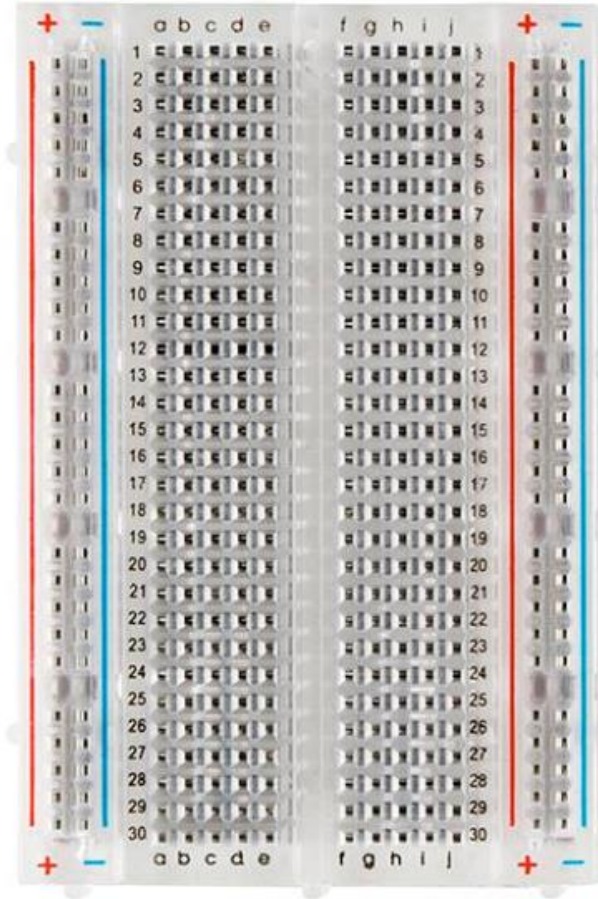
- **Circuit components are physical things!**
- Circuit diagrams are only models of physical phenomena
- Perfect resistors, inductors, and capacitors are only idealized
 - Every capacitor is an inductor if you go high enough in frequency
 - Every inductor is a capacitor if you go high enough in frequency
 - Even simple wires have self-inductance and capacitance
- All components will have typical operating conditions
- Always refer to the datasheets for your components
- **Measure every component before you put it on your circuit**

Lab breadboards

- The red connection points on the breadboards are DC voltage outputs
 - $\pm 15\text{V}$ for op-amp power supplies
 - $+5\text{V}$ for digital logic reference
 - *Do not connect these to the function generator*
- The coaxial BNC connectors are for getting signals onto the board

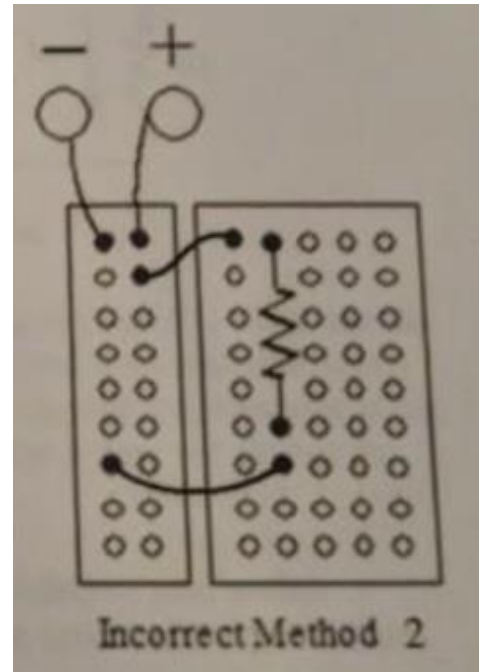
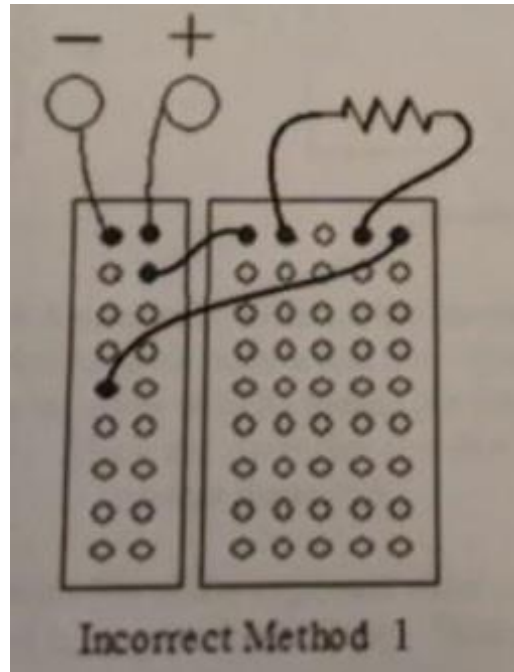
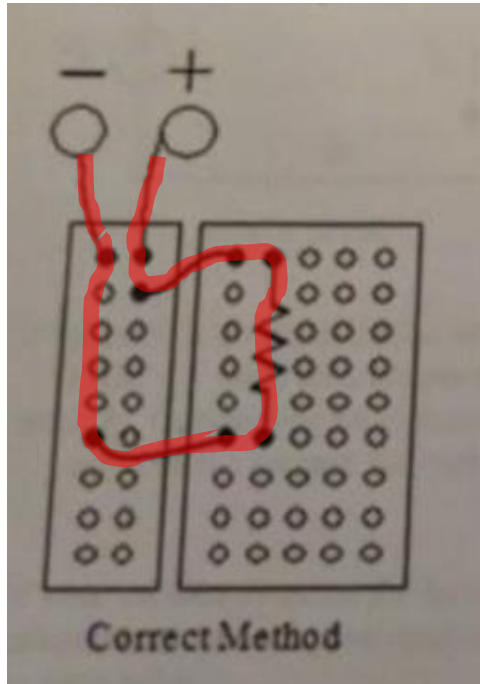


Breadboards

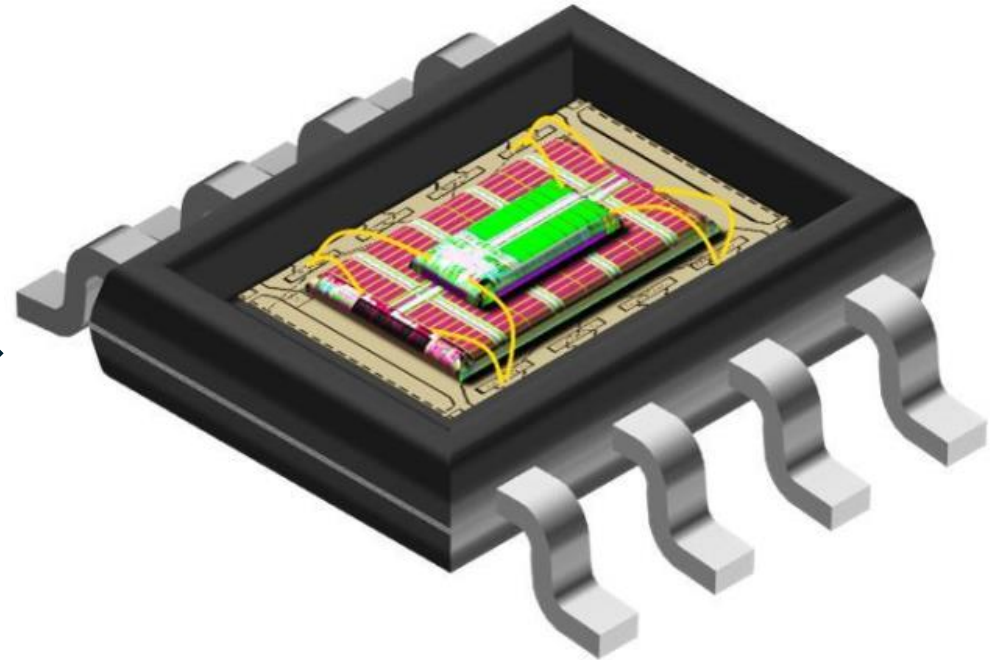
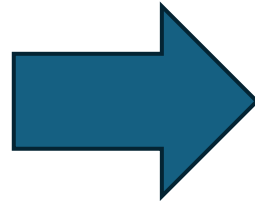
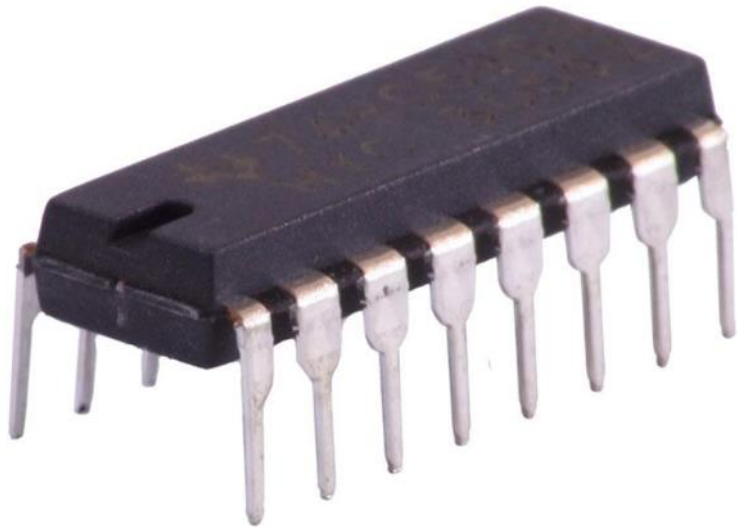


Common breadboard mistakes

Examples of correct and incorrect methods of building Circuit 1 on the breadboard are shown in Figure 8. If you understand why the Incorrect Methods will not give you your expected values for V_{ab} and I_{ab} , and you can avoid these common mistakes, then your breadboard circuits should always work as expected.



Integrated circuits



How to survive the 334 labs

- Read the lab manuals and plan your experiments in advance
- Do theoretical calculations, simulations, etc. before the start of your lab session
- Take *detailed, organized* notes of everything you did, why you did it, and what the result was
- Make sure you know what you are connecting to your circuit
 - Measure your resistors and capacitors
 - Read the part numbers on integrated circuits
 - Measure power supply and function generator outputs
 - *Don't trust the labels on the bins in the lab!*
- Ask for help early if something isn't making sense: sometimes equipment is broken, sometimes lab manuals have typos, sometimes you need a nudge in the right direction.

What to do *this week* regarding labs?

- Read the syllabus
- **Read The Syllabus**
- Install LTSpice and get acquainted with the basics:
<https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>
- Bare minimum: read the manual for the first lab
“Lab A0 – Simulation Intro.pdf”
- Review textbook material in advance, attend lectures, ask questions. Thevenin theorem and AC analysis (e.g. phasor representation of AC signals) are particularly important for the first two labs.
 - WEEK 1 READINGS: chapters 3 to 6 in Storey